

NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

Research Project Report on the Topic:
The relationship between Central Bank Interest Rates and bank stock prices using trading volume as an additional indicator

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Abstract

This paper aims to study the relationship between the Central Bank Interest Rates and the dynamics of bank stock prices, using trading volume as an indicator of the stability of market reaction. The research will analyze the literature related to the topic, systematize the knowledge gained from them, and also consider the application of this knowledge in practice, using data from open sources on the interest rate and the history of the bank stock prices and their trading volume. A model confirming the relationship will also be created.

Аннотация

Данное исследование направлено на изучение взаимосвязи между ключевыми ставками Центральных банков и динамикой курса акций банков, используя объем торгов в качестве показателя стабильности реакции рынка. В ходе исследования будет проанализирована литература, относящаяся к теме, систематизированы знания, полученные из них, а также рассмотрено применение этих знаний на практике, с использованием данных из открытых источников о ключевой ставке и истории стоимости акций банков и объема их продаж. Также будет создана модель, подтверждающая взаимосвязь.

Key words

Trading volume - the total number of units of a financial asset (ex. bank stocks) bought and sold over a specific time, indicating market activity.

Interest rate - the amount of interest due per period (usually per 1 year), as a proportion of the amount, which was borrowed, lent, deposited.

Monetary policy - actions by a central bank to manage the economy stability (ex. adjusting the interest rate)

Policy Interest Rate – the primary interest rate set by a central bank to stabilize economy and control inflation.

Bank stocks – units of a financial asset, shares issued by banks and traded on stock exchanges.

1. Introduction

Trading has become a fairly popular way to earn money. Many people want to predict how stocks will behave in the future. This project aims to find out if it is possible to assume an increase in bank shares based on a change in the policy interest rate. Using the event study method, taking the rate change as an event, mathematical statistics formulas and Python features, I plan to analyze the relationship between the policy interest rate and bank stock prices, also based on trading volume over the past 10 years, using various datasets, as well as create a model explaining this relationship. The conclusion of this study will be relevant for modern traders. This study, based on 10 years of data, will be able to provide a clear overview of market behavior and answer the question of whether changes in the policy interest rate are a determining factor in the growth or decline of bank shares?

2. Methodology

2.1 Data observation

- Data Collection

Source:

<https://www.kaggle.com/datasets/rockyt07/us-banks-stock-all-time> - zip.file of 5 datasets (banks stock price history)

<https://fred.stlouisfed.org/series/FEDFUNDS> - federal fund rate

6 data sets are used, stock price history of 5 US banks + federal fund rate

Time period:

1.01.2015 - 1.01.2025, daily in each dataset

- Variables and Features

In stock price data:

Filtered by date, close price of stocks, volume.

Originally: Date Ticker, Bank Name, Open, High, Low, Close (prices), Volume, Dividends, Stock Splits

In Federal fund data:

Date, interest rate

- Data Quality

No missing (checked by using `df.info()`) on each dataset

- Sample Size

5 banks of US, daily data about stock prices for 10 years in each dataset

interest rate daily data for 10 years

2.2 Hypothesis

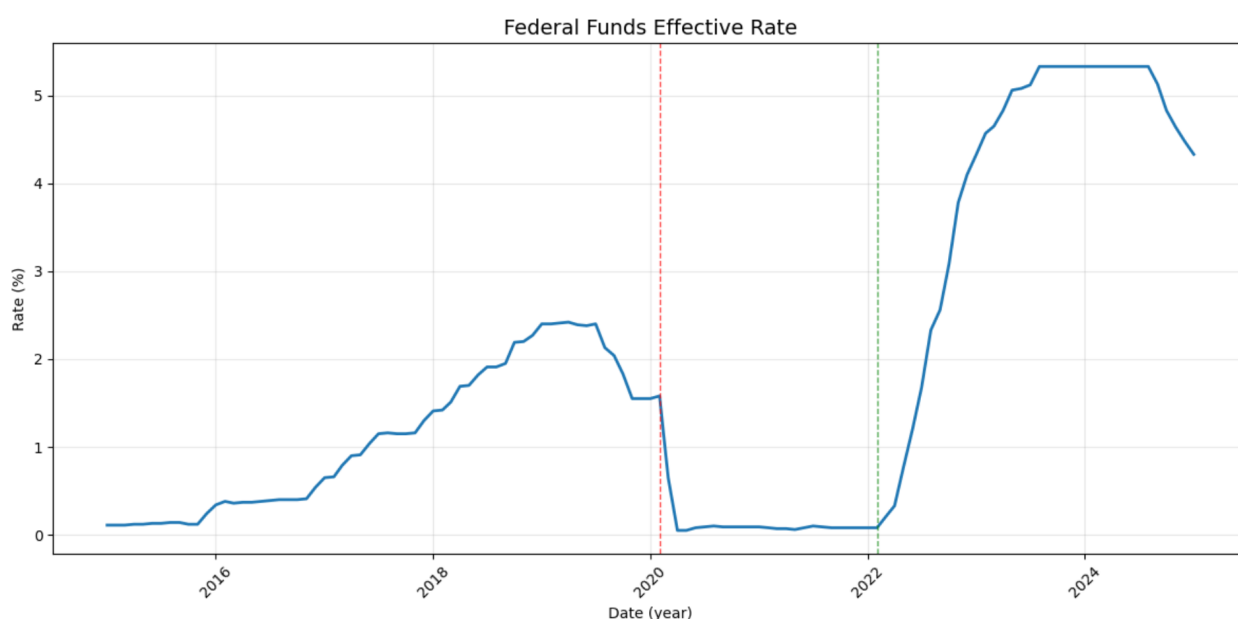
For this study, I propose 3 hypotheses:

- 1) There is a direct relationship between bank stock prices and the federal fund rate, it's one of the main factors to predict bank stock price behavior.
- 2) There is almost no relationship whatsoever; the policy interest rate is not a determining factor.
- 3) A significant relationship can be observed in certain time periods. To study other periods, it is worth considering the events that occurred there separately from others.

Regardless of which hypothesis turns out to be correct, this knowledge can help novice investors who are planning to invest in bank shares.

2.3 Visualization and relationships of data

This study examines the policy interest rate in the United States (Federal Funds rate). Using its dynamics as an example, we can trace how it responds to serious crises (ex. Covid-19 period of 2020-2022). This simple observation makes it clear that any analysis of the policy interest rate over the past 10 years will inevitably encounter large interest rate fluctuations.



Federal Funds Effective rate data from 01.01.2015 to 01.01.2025. There has been a noticeable decline during Covid-19 period.

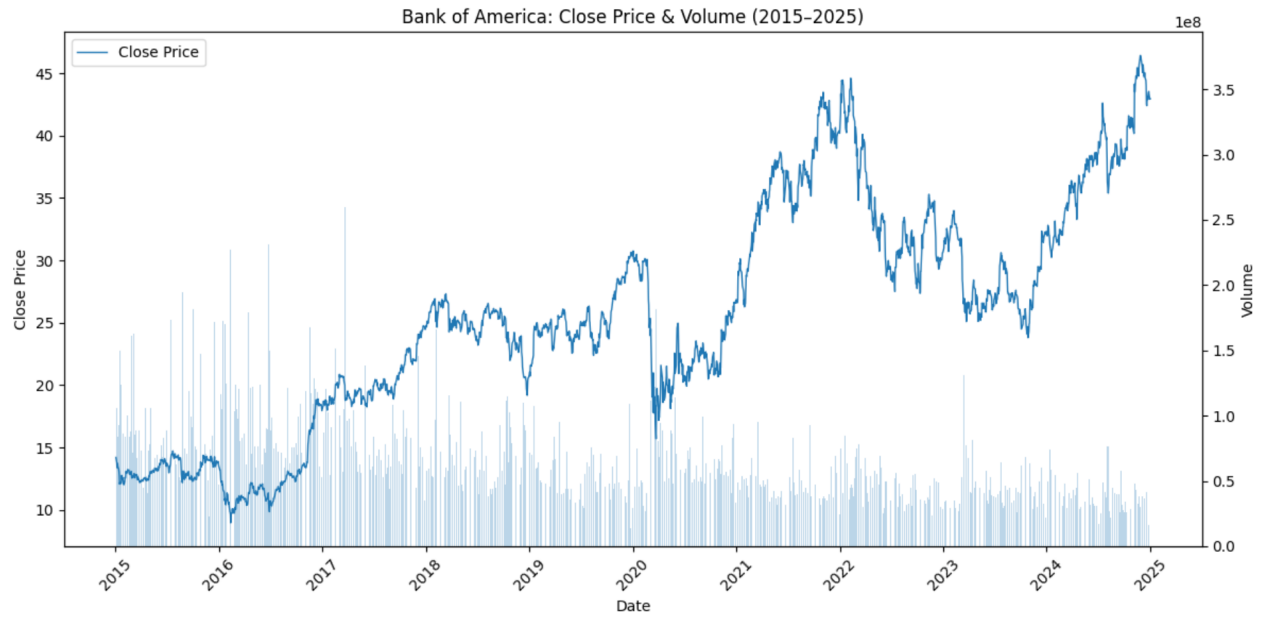
Some trends that will influence the relationship are visible:

- Steady growth until 07/2019
- Fast drop after 02/2020
- Stable from 04/2020 to 02/2022
- A sharp increase after 02/2022

That dynamic could have created some issues. For example, Ben S. Bernanke and Kenneth N. Kuttner in their research “What Explains the Stock Market’s Reaction to Federal Reserve Policy?” concluded that the market reacts fairly strongly to surprise funds rate changes, but it reacts little, if at all, to the component of funds rate changes that are anticipated by futures market participants. And Järvinen, K. in his research “The impact of interest rates on bank stock

returns” finds a paradox that, according to previous studies, rising policy interest rates negatively affect bank stocks, although theoretically they should increase their profitability, but according to the results obtained, changes in rates have a positive effect on the profitability of bank shares. Stocks data was studied for the period 2012-2021. He explains the unexpected results by the difference in the chosen period and their peculiarities. This example clearly shows that the results of policy interest rate dynamic studies may vary depending on the period they studied.

The banks stocks data can be represented as a simple graph (one for each bank).



Example, Bank of America close price and volume data, sorted by date.

Using this dataset for an example of visualization and correlation calculation with data from Federal Fund dataset. Calculations were performed using the Pearson correlation formula ¹with the stats module² in Python.

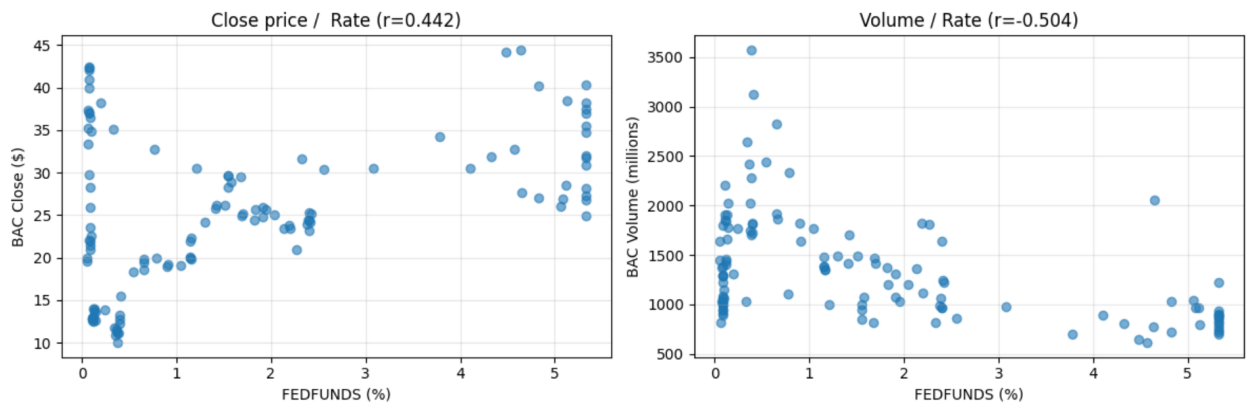
$$r_{xy} = \frac{\sum (x_i - M_x) (y_i - M_y)}{\sqrt{\sum (x_i - M_x)^2 \cdot (y_i - M_y)^2}}$$

Python code:

```
corr_close, p_close = stats.pearsonr(monthly_data['Close'], monthly_data['FEDFUNDS'])
```

¹ A mathematical formula which is using to calculate correlation coefficient, which measures the linear correlation between two sets of data.

² The scipy package in Python contains various toolboxes dedicated to common issues in scientific computing (ex. statistics).



The result of those calculations, correlation between:

Close price and Federal fund rate: **$r = 0.4421$** , **$p\text{-value} = 0.0000$** (.4f for values)

Stock volume and Federal fund rate: **$r = -0.5044$** , **$p\text{-value} = 0.0000$** (.4f for values)

Where $r = \text{correlation}$

Based on Chaddock scale³, correlation between 0.3 and 0.5 is moderate positive. As for the second result, it is called negative evident for -0.5 to -0.7.

As I already mentioned, you can visually see how both graphs (stock data and federal funds) have changed their trends significantly since 2020, so I suggest looking at the statistics for different periods. For this section, I suggest determining:

- 1) Stable period (1.01.2015;1.01.2020) - direct upward trend can be seen
- 2) Covid19 period (1.01.2020; 1.01.2022) - economics reaction to that issue
- 3) Post-covid period (1.01.2022; 1.01.2025) - “healing” of the economy

It's hard to say that Covid19 is the cause of the sharp fluctuations in the charts (it's impossible to take into account all the reasons for this), but visually you can tell how much it has affected the economy. This will simply be a designation of periods where sharp changes in trends are visually noted.

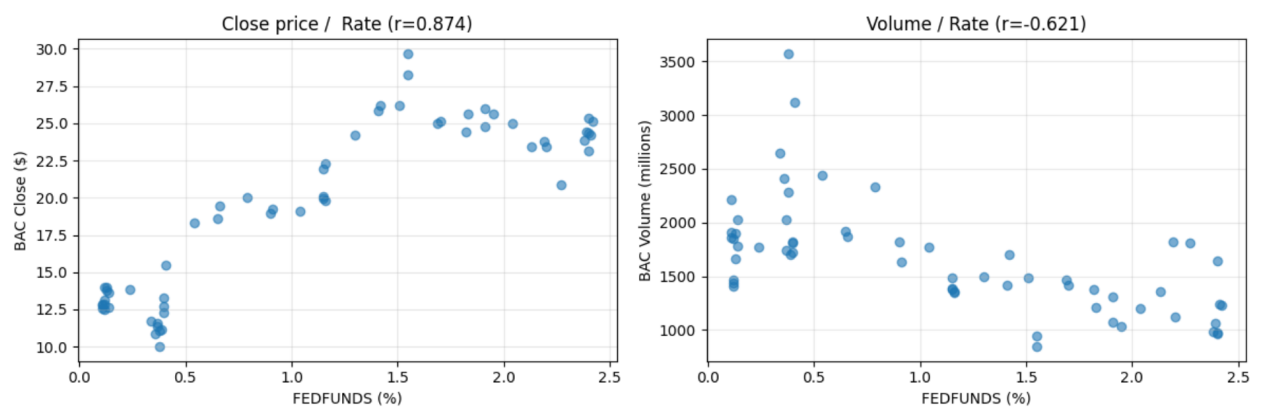
The results for those periods:

	Stable period	Covid19 period	Post-covid period	SUM
Close price and Federal fund rate	$r=0.8744$ $p\text{-value}=0.0000$	$r=-0.0853$ $p=0.6918$	$r=-0.2416$, $p\text{-value}=0.1558$	$r=0.4421$ $p\text{-value}=0.0000$

³ A statistical method that helps to define any dependency between variables and its strength using correlation coefficients.

Stock volume and Federal fund rate	$r=-0.6212$ $p\text{-value}=0.0000$	$r=0.0794$ $p\text{-value}=0.7123$	$r=-0.2604$, $p\text{-value}=0.1250$	$r=-0.5044$ $p\text{-value}=0.0000$
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Correlation results are totally different, but before observing them, we need to focus on p-value. P-value is the probability of being wrong about the existence of a relationship or difference. In other words, it's the chance that the result is due to chance. In period 1), p-value = 0 (.4f), the relation is confirmed ($p\text{-value} < 0.05$). The result is significant. But for period 2) and 3), p-value is pretty big, that means, there is almost zero chance, that there is a correlation between two groups. It turns out that to identify a direct correlation, it's worth considering the period 2015-2020. But to examine the period after 2020, it's worth reviewing the events that influenced Central bank decisions and market reactions.

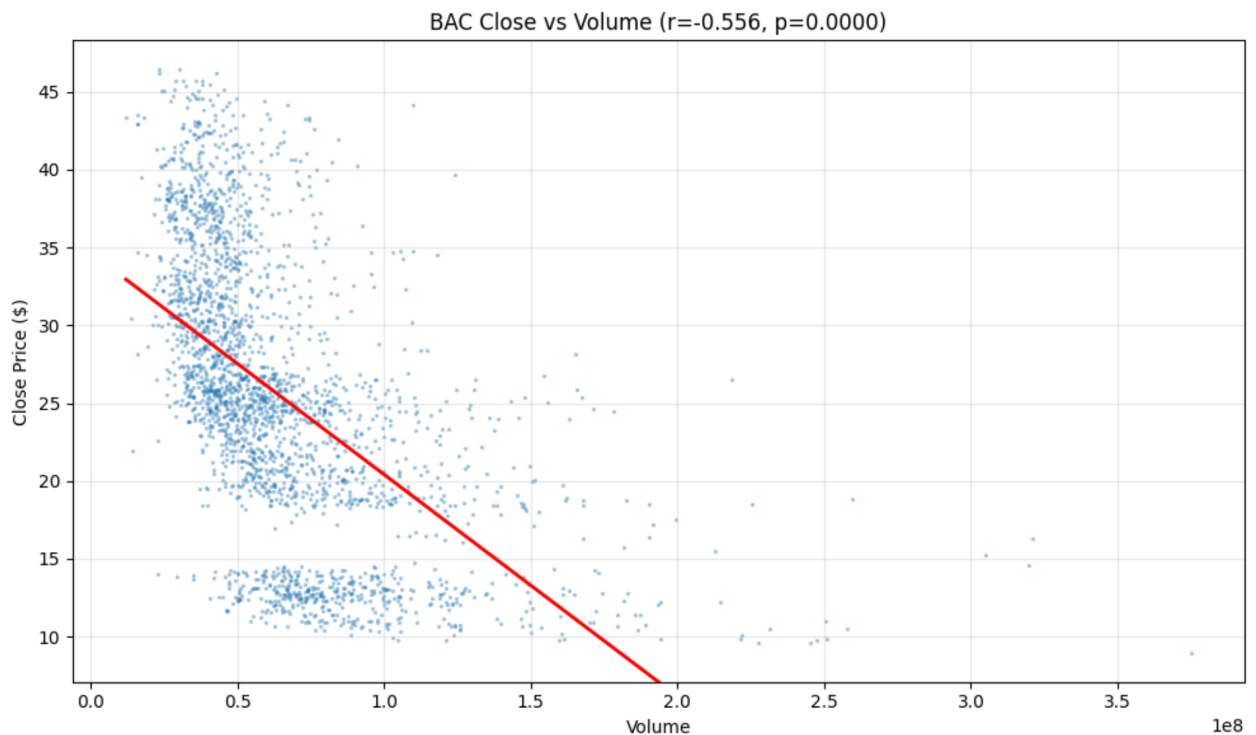


Data of period 1), the only significant one (by p-value).

The correlation between stock price and interest rate is high (>0.8), which means there is a true relation. As for the trading volume, it's negative evident correlation. The negative correlation can be explained by the fact that in an unstable economic situation (the policy interest rate is rising), people prefer to invest their savings in deposits at a fixed rate rather than try their luck on the stock market.

I would like to highlight the fact that the trading volume does not directly correlate with the stock price. These calculations show that there is a connection, but trading volume is not considered an important indicator for predicting stock dynamic with $r = -0.5559$ and $p\text{-value} =$

0.0000 for time period 1.01.2015 - 1.01.2025 (pretty same result for overs periods)



Experienced investors point to volume as a significant indicator, but its connection to price is not direct and depends on the current trend in stock growth. They indicate several possible consequences of changing volume depending on price.

Volume Stock price	Up	Down	Stable
Up	A signal for short and medium-term investments.	Declining volume and rising price indicate that the uptrend is ending.	When volume remains relatively stable and the price keeps rising, the uptrend is stabilizing.
Down	The price will go upward soon.	Decreasing volume in parallel with falling price indicates a downtrend.	Volume stops decreasing, and the price keeps going down.
Stable	If this occurs in a downtrend, it means that the trend is weakening, whereas in an upward or sideways trend, it suggests the trend is likely to continue.	The price will decrease soon.	-

3. ML modelling

A supervised machine learning model was used to analyze the relationship between the federal funds rate and bank stock prices. Specifically, a linear regression model was used to estimate the relationship between the interest rate (the independent variable) and stock prices (the dependent variable). Also, it was chosen for its suitability for economic analysis, as it allows shows an impact of interest rate changes on stock prices.

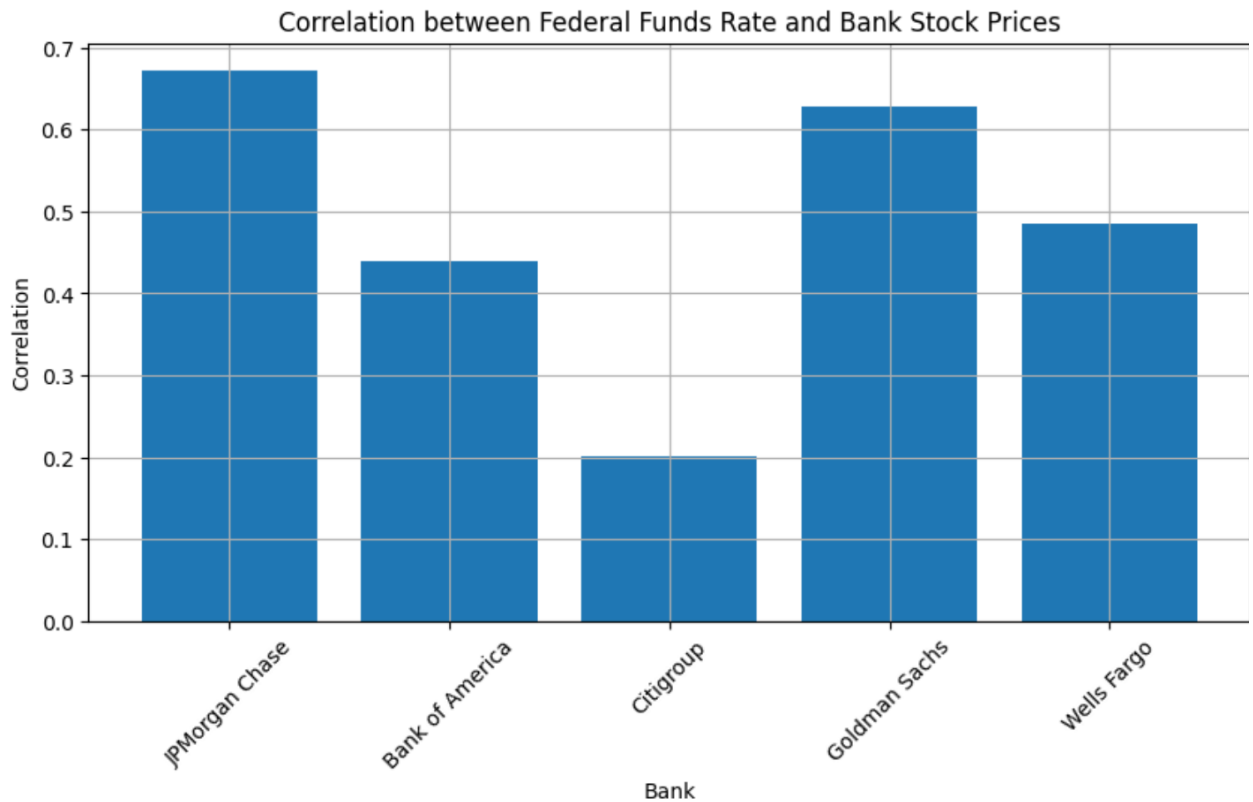
The model assumes a linear relationship of the following form:

$$\text{Price} = (a + b) * \text{Rate}$$

where Price is the stock price, Rate is the federal funds rate, a is the intercept, and b is the slope.

The model was trained using historical data of stock prices and Central Bank rate for the period under study (1.01.2015 - 1.01.2025). The slope coefficient indicates the stock price changes with a one percentage point increase in the interest rate (change per 1% rate). Several performance metrics were used to evaluate the model, including the *coefficient of determination* (R^2), *mean absolute error* (MAE), and *root mean square error* (RMSE). In addition, a correlation coefficient was calculated to measure the strength and direction of the relationship between variables.

The graph and the table below present the results of the linear regression model for each bank in a period of 2015 - 2025.



Bank	Correlation	Intercept	Slope	R2	MAE	RMSE
JPMorgan Chase	0.6719	75.5161	16.102	0.4515	27.3942	33.1077
Bank of America	0.4399	21.7827	2.089	0.1935	6.1803	7.9558
Citigroup	0.201	46.362	1.0055	0.0404	7.9134	9.1424
Goldman Sachs	0.6272	184.2421	33.8092	0.3934	60.996	78.3212
Wells Fargo	0.4849	37.3898	2.182	0.2352	5.5882	7.3407

However, based on the findings from the previous chapter and my classification of periods, I propose analyzing them separately.

○ **Stable period (1.01.2015;1.01.2020)**

Bank	Correlation	Intercept	Slope	R2	MAE	RMSE
JPMorgan Chase	0.8903	47.0991	22.1918	0.7927	6.8156	9.2611
Bank of America	0.8711	12.4417	6.1187	0.7588	2.2342	2.8147
Citigroup	0.7470	37.5365	8.0064	0.5580	4.5683	5.8145
Goldman Sachs	0.4662	155.2309	15.3729	0.2174	20.2571	23.8001
Wells Fargo	0.3023	40.1392	1.2740	0.0914	2.6653	3.2775

During that period, a strong positive correlation is observed between the interest rate and bank stock prices. The most pronounced relationship is observed for JPMorgan Chase and Bank

of America (almost 0.9), and R^2 values are almost equal 0.8, demonstrating the model's strong explanatory power. **Positive slope values for all banks indicate that interest rate increases were accompanied by rising stock prices.** But for some banks (ex. Wells Fargo) the relationship is weaker, as evidenced by low R^2 values.

○ **Covid19 period (1.01.2020; 1.01.2022)**

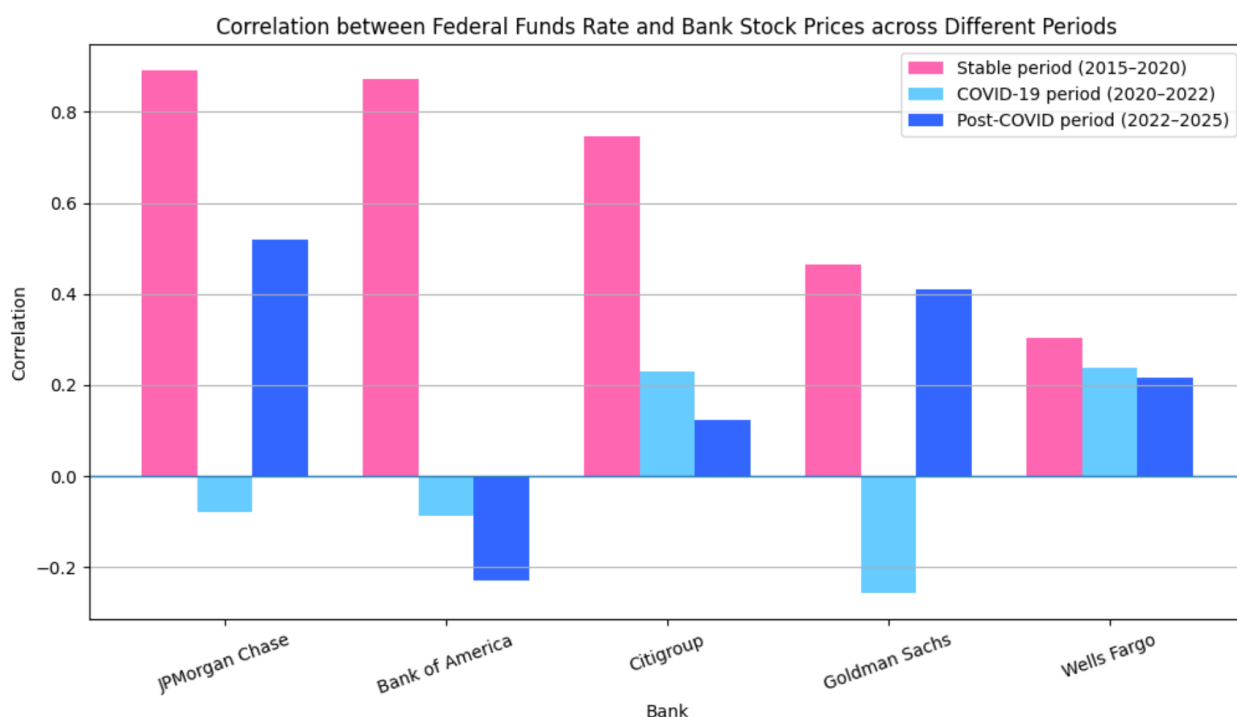
Bank	Correlation	Intercept	Slope	R2	MAE	RMSE
JPMorgan Chase	-0.0793	116.6248	-5.0351	0.0063	23.8041	25.9650
Bank of America	-0.0885	29.9075	-1.6331	0.0078	6.8371	7.5420
Citigroup	0.2292	50.1683	5.8339	0.0525	8.9315	10.1684
Goldman Sachs	-0.2575	264.5579	-49.2162	0.0663	69.0510	75.7811
Wells Fargo	0.2372	31.7074	5.2599	0.0562	8.0618	8.8421

During the COVID-19 period, there has been an instability in the relationship. This is explained by the fact that during the pandemic, the stock market was influenced primarily by external factors, which significantly reduced the importance of the interest rate in price formation.

○ **Post-covid period (1.01.2022; 1.01.2025)**

Bank	Correlation	Intercept	Slope	R2	MAE	RMSE
JPMorgan Chase	0.5188	107.4959	11.0115	0.2691	28.1081	32.8039
Bank of America	-0.2293	36.0924	-0.6978	0.0526	4.5214	5.3555
Citigroup	0.1240	46.5163	0.6014	0.0154	7.5027	8.7000
Goldman Sachs	0.4094	286.6362	18.3053	0.1676	57.1431	73.7498
Wells Fargo	0.2153	42.0930	1.1031	0.0463	7.5097	9.0473

In the post-COVID period, there has been a partial recovery in the relationship between interest rates and bank stock prices, but the correlation usually remains weak. For some banks, (JPMorgan Chase and Goldman Sachs) a positive correlation is visible, indicating a return to the influence of interest rates on stock prices. However, for other banks, the relationship remains weak or even negative, and R^2 values remain low, suggesting that interest rates are not really important now. This suggests that the economy is in a recovery phase similar to that seen in 2015-2020, but the influence of interest rate policy has not become a dominant factor yet.



4. Conclusion based on gained data

This study tested three hypotheses about the nature of the relationship between Central Bank Interest Rates and bank stock prices.

- 1) There is a direct relationship between bank stock prices and the federal fund rate.
- 2) There is almost no relationship.
- 3) A significant relationship can be observed in certain time periods.

The **first hypothesis**, which assumes a direct correlation between the interest rate and bank stock prices, is **not supported**. During the stable period (2015–2020), a positive correlation is indeed observed, indicating a significant influence of the interest rate on bank stock prices. However, during the crisis period, the correlation is virtually nonexistent.

The **second hypothesis**, which asserts the absence of a significant correlation, is **not supported**. During the COVID-19 pandemic (2020–2022), the correlation between the variables is virtually nonexistent, with coefficients near zero, but it is clearly visible during stable periods. This indicates that during a crisis, the key rate ceases to play a decisive role, yielding influence to other factors during crises.

The **third hypothesis** is **supported**. The results show that the relationship between interest rates and bank stock prices is variable and significantly dependent on the economic situation. In the post-COVID period (2022–2025), a partial restoration of the relationship is observed, but it remains less pronounced than during the stable period. This suggests that Central Bank Interest Rates are an important, but not the only, factor influencing bank stock prices. Their influence

increases during periods of economic stability and weakens during crises and heightened uncertainty.

These results may be useful for beginning investors, as they demonstrate that when making investment decisions, it is important to consider not only the interest rate level but also the economic situation.

5. Literature review

5.1 Karpoff, J. M. (1987). The Relation Between Price Changes and Trading Volume: A Survey. The Journal of Financial and Quantitative Analysis, 22(1), pp. 109–126.

Description

The article reviews, critically analyzes and complements some previous researches on the relation between price changes and trading volume. The author reveals a pattern: volume is positively related to the magnitude of the price change and, in equity markets, to the price change per some period of time. He explains the results of previous researches and provides information about several attempts of making a model that explains the relationship between price changes and trading volumes. The author assumes that observations of simultaneous large volumes and large price changes can be traced to their common ties to information flows.

Connection to the topic

In this article, author confirms the relationship between trading volume and stock prices, analyzing previously known information. The part about information flows, can be used to assume, that the change of Policy Interest Rate will cause an information flow, forcing the market to response, therefore the stock prices and the trading volume will change.

5.2 Bernanke, B. S., & Blinder, A. S. (1992). The Federal Funds Rate and the Channels of Monetary Transmission. *The American Economic Review*, 82(4), 901–921.

Description

The article asserts, that monetary policy works at least in part through bank loans and dynamic response of the economy to innovations in the policy interest rate will reflect the true structural impact of monetary policy. The author presents the responses of bank balance sheet variables react to a federal funds rate shock. He also mentioned that short-run variations in the policy interest rate are mostly attributable to Federal Reserve policy decisions, not to fluctuations in the demand for reserves, The author's results are consistent with the view that monetary policy works in part by affecting the composition of c, as he directly stated in the paper.

Connection to the topic

In the paper, the author examines the impact of the policy interest rate on bank balance sheet variables. This influence, proven in the study, is directly related to the topic of the project, especially bank stocks, which, obviously, are also changing in that situation. The author's claim about short-run variations in the policy interest rate, proved in the study, is reflected in the consideration of Federal Reserve policy decisions as important event for the change of bank stock prices. The conclusion basically confirms the relationship between bank assets and the change of policy interest rate.

5.3 Bernanke, B. S., & Kuttner, K. N. (2005). What Explains the Stock Market's Reaction to Federal Reserve Policy? The Journal of Finance, 60(3), 1221–1257.

Description

This paper analyzes the impact of changes in monetary policy on equity prices. Authors say that the change of policy interest rate actions on expected excess returns account for the largest part of the response of stock prices. They point to the surprise element of any specific change in the Federal Funds as the reason for the stock market's reaction. One of their conclusions was to confirm the existence of a strong 1-day reaction of the stock market to unanticipated changes in Federal funds rate, receiving up to 80% confidence intervals associated with the estimated industry responsibilities. This study has documented a strong response of the stock market to unexpected monetary policy actions, using Federal funds futures data from different periods of time.

Connection to the topic

The authors' confirmation of the stock market's rapid reaction to monetary policy actions may be the basis for conclusions about changes of bank stock prices. Information from that paper may be an important part of the study of the relationship between Central Bank Policy Rates and bank stock prices. This study also raises the question: is the change in the price of bank stocks different from stocks in other growths?

5.4 Järvinen, K. The impact of interest rates on bank stock returns **[Master's thesis, University of Vaasa]. Vaasa: School of** **Accounting and Finance.**

Description

This paper examines the impact of interest rate changes on the profitability of bank stocks in a low interest rate environment. The author finds a paradox that, according to previous studies, rising interest rates negatively affect bank stocks, although theoretically they should increase their profitability, but according to the results obtained, changes in rates have a positive effect on the profitability of bank shares. Stocks data was studied for the period 2012-2021. He explains the unexpected results by the difference in the chosen period and their peculiarities. He also uses models and data visualization in his work.

Connection to the topic

The topics of both studies are very similar. It is interesting how the author obtains data that contradicts the theory of previous works. This suggests that this study will have similar problems that the author faced. Similar values are used in calculations and the formation of relationships. In addition, the time period is almost identical.

5.5 Sohn, Wook; Sung, Byeongmook Research Report The Effects of **Central Banks' Rate Change Patterns on Financial Market Variables KDI** **Policy Study, No. 2013-01 Provided in Cooperation with: Korea** **Development Institute (KDI), Sejong**

Description

This article discusses the fact that an increase in interest rates, as a rule, has a negative impact on stock returns, but various research results have been obtained throughout the history of the stock market. The focus of the research is on the crisis of 2008. The results show that the nature of interest rate changes, such as range and frequency, has a significant impact on important financial variables in the monetary policy transmission mechanism. In this study, the author studies terminology in depth and uses mathematical analysis to obtain a result. The final conclusion is that economic theory, which includes the behavior of central banks and patterns of interest rate changes, requires further development.

Connection to the topic

This paper explores a similar topic, coming to an unexpected conclusion. In the end, the author leaves questions that require additional research.

5.6 Rigobon, R., & Sack, B. (2001). Measuring the Reaction of Monetary Policy to the Stock Market (Working Paper No. 8350). National Bureau of Economic Research.

Description

In this paper, to identify the reaction of monetary policy to the stock market. It is difficult to assess this reaction, because not only the Fed can influence the market, but also changes in the market can prompt the Fed to react. The author uses complex models and many ways to visualize dates to make calculations as accurate and visual as possible. The Fed really reacts significantly to stock market movements, which is confirmed by the arguments of the author in his paper. This article claims that the stock market has become the third important factor.

Connection to the topic

This article examines the reasons for the Fed's behavior, finding a connection in the stock market, though not a primary one. This conclusion can confirm the relationship between Central Bank Interest Rates and bank stock prices which is the subject of my research.

5.7 Cochrane, J. H., & Piazzesi, M. The Fed and Interest Rates: A HighFrequency Identification (Working Paper No. 8839). National Bureau of Economic Research

Description

This article examines monetary policy shocks determined based on changes in the Fed's funds targets relative to daily interest rate data. The author argues that the Fed is regulating the changing situation, rather than gaining momentum. As is often the case, the greater the shock, the more unusual the reaction. Interest rates are moving in the same direction as political shocks, and on a surprisingly large scale.

Connection to the topic

The paper makes sure that long-term rates react strongly to unexpected actions by the Fed. Also, the banking business (the central unit in the topic "The relationship between Central Bank Interest Rates and bank stock prices using trading volume as an additional indicator"). It is based on a percentage margin. A strong and unexpected shift in rates directly affects the profitability of banks, and hence their shares. This connects the topics of the two studies

6. Problem:

Insufficient study of the relationship between changes in the central bank's policy rates and the reaction of bank stocks, using trading volume as an indicator of market reactions.

7. Plan:

1/2026	1. Analysis of the impact of monetary policy on bank stocks using various studies 2. Analyze the role of trading volume as an indicator of market reactions using various studies 3. Collect data and prepare the dataset
2/2026	4. Visualize the data 5. Correlation analysis 6. Event study around Federal Reserve policy decisions 7. Analysis of asymmetric reactions
3/2026	8. Develop a conceptual model of the relationship of the studied variables 9. Interpret the results obtained in the context of existing theories
4/2026	10. Correct possible inaccuracies in the finished work